Environmental Inequalities, Race, and Class: Mapping the Industrial Landscape of Mid-Century American Cities

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This presentation showcases “Fires of Industry: Environmental Inequalities in Mid-Century America,” a public digital humanities project developing, mapping, and visualizing a new dataset of environmentally burdensome sites across dozens of American cities circa 1950. By juxtaposing this environmental data with racial demographics and median income, the project explores historical environmental disparities and their ongoing impact on health inequalities.

“Fires of Industry” is part of a National Institutes of Health-funded project, The Impact of Redlining and Place-Based Systemic Racism on Health Inequalities at Mid-Life. While that project examines the impact of redlining on contemporary health disparities, it aims to move beyond the widespread reliance on redlining spatial data in studies of racial and class disparities, especially those related to the environment (e.g., urban heat islands (Hoffman et al. 2020), air pollution (Cushing et al. 2023), bird biodiversity (Ellis-Soto et al. 2023)) and health (e.g., asthma (Nardone et al. 2020), preterm births (Krieger et al. 2020), food access (Shaker et al. 2023), and gun violence (Benns et al. 2020)).

Numerous studies have used the spatial dataset developed through Mapping Inequality: Redlining in New Deal America (Nelson et al. 2016) to investigate these relationships. While this work has generated important insights, the causal link between 1930s mortgage discrimination and contemporary environmental and health disparities remains contested, raising the need for new historical spatial datasets like Fires of Industry to better understand the origins of these inequalities.

In the past few years, numerous studies on contemporary inequalities have analyzed the spatial dataset from redlining "security maps" developed by the Home Owners’ Loan Corporation (HOLC) in the 1930s, which collaborators and I developed and shared through the digital humanities project Mapping Inequality: Redlining in New Deal America (Nelson et al. 2016). While this research has brought welcome and much-needed attention to the connection between government-endorsed spatial and structural racism in the twentieth century and twenty-first-century environmental and health inequities, it has also faced criticism for the tenuous causal link between mortgage discrimination in the 1930s and today’s inequalities. Robert Gioielli (2022) aptly calls this the "redlining vampire": treating redlining as represented in the HOLC maps as an unchanging phenomenon that is evoked rather than historicized. He and others argue that "to focus on the maps and to use them as the primary data set in our analysis of why a particular neighborhood is too hot, too poor, and too crime ridden, is … a scholarly miscalculation that fails to get us close to an accurate picture of the causes of contemporary racial inequalities" (Gioielli 2022; Mallach 2024; Graetz / Esposito 2022). Fires of Industry is, in part, an effort to address this critique and advance research across disciplines on the history of spatial racism and its legacies through the creation and analysis of a spatial dataset that has a more direct causal relationship to environmental and health inequalities.

One major reason that many researchers have used the 'Mapping Inequality' redlining dataset is the lack of alternatives. No detailed datasets of the environmental landscape of American cities exist prior to the government's efforts after establishing the EPA in 1970. “Fires of Industry” aims to fill this gap by creating a multi-city, granular dataset, developing an interactive mapping and visualization tool for students and citizens, and sharing the data for researchers to use and build upon.

We are developing this dataset using Sanborn Insurance maps published around 1950, which, as Figure 1 shows, can dramatically illustrate the proximity of environmental burdens (in this case a gasometer, dry cleaner, dump, and coal and junk yards) and African American institutions, two of which served particularly vulnerable populations of the young and the old: Carver Elementary School, the Afro American Old Folks Home, and a Black Baptist Church.

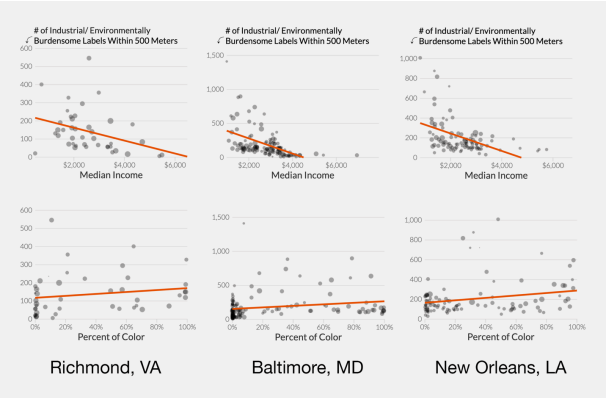

Figure 1. Stitched-together maps from Insurance Maps of Richmond, Virginia, Volume Three (New York: Sanborn Company, 1952) with overlay to draw attention to industrial sites and African American institutions.

We have georeferenced nearly 17,000 individual Sanborn maps across two dozen cities to date, with a goal of reaching at least 25,000 maps for forty cities. Together these maps present a highly detailed portrait of the built environment, particularly the industrial landscape. We are using the MapReader software to recognize and extract geocoded labels on the maps (Hosseini et al., 2023). We have developed a dictionary of analytically relevant terms that we use to filter the output. For example, oil, fertilizer, and abattoir are categorized as "environmentally burdensome"; D (dwelling) and apts as "residential"; store and office as "commercial"; etc.

This spatial data allows us to visualize and analyze the industrial and environmental landscape of dozens of cities. For example, the figure below shows the environmentally burdensome points we have identified in Richmond, Virginia, visualized as a heatmap (using the default kernel-density algorithm in the deck.gl library) and overlaid on data of the African American population and median incomes from the 1950s census, showing in detail that poorer neighborhoods with predominantly Black and brown residents were more likely to live near environmental hazards.

Figure 2. Screenshot of "Fires of Industry" showing "Environmentally Burdensome" labels heatmap, African American-majority areas, and median income in Richmond, VA

In "Fires of Industry," we also analyze this data statistically. Linear regressions are visualized as trendlines to illustrate in another way how the burdens of industrialization and environmental inequality fell on communities of color and working-class neighborhoods.

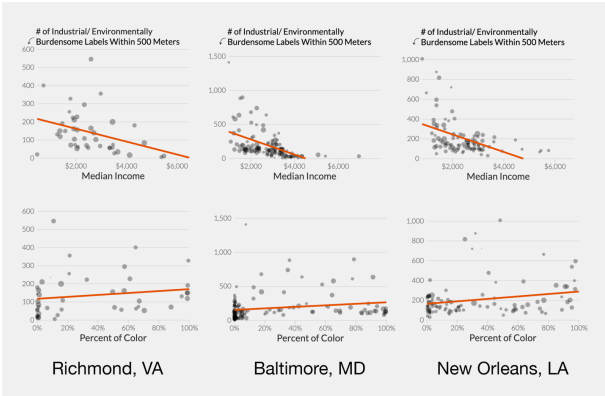

Figure 3. Patterns in median income and racial demography by census tract for three cities.

In showcasing this well-developed but still work-in-progress project, this presentation will connect directly with the conference themes of accessibility and citizenship. "Mapping Inequality" aimed to make the HOLC redlining maps and their derived spatial data widely accessible. While it has unquestionably been successful as a public digital humanities project, from the perspective of many humanists, its data have been overused and misused in environmental and health studies, raising important questions about the opportunities and challenges of transdisciplinary research. "Mapping Inequality" has also connected with citizenship more than we anticipated, being used by advocacy organizations and some policymakers to address historical and ongoing racial and class inequalities, for example in state and city studies of reparations. We hope "Fires of Industry" will further this work by creating a freely accessible dataset for researchers across disciplines and an interactive, data-rich mapping application that can be used by citizens striving for a more equitable future.

Bibliography

Benns, Matthew / et al. (2020): "The Impact of Historical Racism on Modern Gun Violence: Redlining in the City of Louisville, KY", in: *Injury* 51, 10: 2192–2198.

Cushing, Lara J. / et al. (2023): "Historical Red-Lining Is Associated with Fossil Fuel Power Plant Siting and Present-Day Inequalities in Air Pollutant Emissions", in: *Nature Energy* 8, 1: 52–61.

Ellis-Soto, Diego / Chapman, Melissa / Locke, Dexter H. (2023): "Historical Redlining Is Associated with Increasing Geographical Disparities in Bird Biodiversity Sampling in the United States", in: *Nature Human Behaviour* 7, 11: 1869–1877.

Gioielli, Robert (2022): "The Tyranny of the Map: Rethinking Redlining", in: *The Metropole* (blog), November 3. https://themetropole.blog/2022/11/03/the-tyranny-of-the-map-rethinking-redlining/ (last accessed April 25, 2025).

Graetz, Nick / Esposito, Michael (2022): "Historical Redlining and Contemporary Racial Disparities in Neighborhood Life Expectancy", in: *Social Forces* 102, published online October 31. https://doi.org/10.1093/sf/soac114 (last accessed April 25, 2025).

Hoffman, Jeremy S. / et al. (2020): "The Effects of Historical Housing Policies on Resident Exposure to Intra-Urban Heat: A Study of 108 US Urban Areas", in: *Climate* 8, 1.

Hosseini, Kasra / et al. (2023): "Living-with-machines/mapreader: End of Lwm", in: *Zenodo*, July 27. https://doi.org/10.5281/zenodo.8189653 (last accessed April 25, 2025).

Krieger, Nancy / et al. (2020): "Structural Racism, Historical Redlining, and Risk of Preterm Birth in New York City, 2013–2017", in: *American Journal of Public Health* 110, 7: 1046–1053.

Mallach, Alan (2024): "Shifting the Redlining Paradigm: The Home Owners' Loan Corporation Maps and the Construction of Urban Racial Inequality", in: *Housing Policy Debate*, published online March 11.

Nardone, Anthony / et al. (2020): "Associations between Historical Residential Redlining and Current Age-Adjusted Rates of Emergency Department Visits Due to Asthma across Eight Cities in California: An Ecological Study", in: *The Lancet Planetary Health* 4, 1: e24–e31.

Nelson, Robert K. / et al. (2016): "Mapping Inequality: Redlining in New Deal America", ed. Robert K. Nelson, *American Panorama: An Atlas of United States History*. https://dsl.richmond.edu/panorama/redlining (last accessed April 25, 2025).

Shaker, Yasamin / et al. (2023): "Redlining, Racism and Food Access in US Urban Cores", in: *Agriculture & Human Values* 40, 1: 101–112.